
TinyAudit: On-Device, Uncertainty-Aware Fairness Audits that Survive (and Expose) Compression

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Abstract

Machine learning is moving onto microcontroller-class hardware, where models are aggressively compressed (pruned, quantized) to fit. Responsible-AI audits, in contrast, are almost always run on full-precision models on a workstation, and they typically look at fairness, uncertainty, and explainability in isolation. We present TinyAudit, an auditing pipeline that measures all three together on the compressed artifact rather than the original, and that profiles its own cost. Theory already tells us calibration and parity cannot generally agree, and that compression harms underrepresented groups; we claim neither as new. Our contribution is measuring what those results do to a deployed model, reported as means over 10 seeds. First, the two lenses do not merely disagree, they select different groups: the ordering by selection rate and the ordering by per-group calibration error cross on UCI Adult and again on COMPAS, whose favorable direction is inverted. Second, and more actionable, compression moves the two in opposite directions: pruning an MLP to 90% sparsity drives its demographic-parity difference down roughly 12-fold while per-group calibration error climbs roughly 10-fold, so a parity-only audit run after compression ranks the most degraded model in the sweep as the fairest, and the same collapse replicates on a third dataset. Third, the uncertainty view is not redundant with calibration: predictive entropy ranks the groups differently from per-group calibration error (mean $|\rho| = 0.21$ over the race cells), so both are needed. Finally, the audit’s own peak working set is about 0.6 MB plus 2.3 KB per sample, constant in test-set size when streamed. That constant does not yet fit the 32 to 512 KB parts at the low end of the TinyML range, and we report the gap rather than claim it closed.

1 Introduction

TinyML runs inference on microcontrollers with 32 to 512 KB of SRAM [19]. To fit that budget, deployed models are pruned and quantized. The auditing tools that would check those models for fairness were built for a different world: cloud notebooks, full-precision weights, and demographic labels available at test time [14, 2, 8]. Existing TinyML benchmarks measure latency, accuracy, and energy, but not fairness or explainability [19]. The result is a gap. The models most likely to be compressed are the least likely to be audited, and when they are audited, the audit sees a model that is not the one deployed.

A second problem is that fairness audits usually stop at point predictions. A model can look fair on selection rates while its uncertainty is biased against a group [13], and abstaining on high-uncertainty inputs does not by itself repair group imbalance [17]. Uncertainty quantification now fits on KB-sized devices [7], but no existing system evaluates those estimates through a fairness lens, and no fairness toolkit fits on the device.

37 TinyAudit closes both gaps at once: a seven-stage pipeline that measures fairness, uncertainty, and
 38 explainability on the compressed model, inside a memory envelope sized for the device being au-
 39 dited.

40 **Relation to known results.** Both results have theoretical ancestors and we want to be precise about
 41 what is new. That calibration and parity-style criteria cannot generally hold at once is established
 42 [11, 16], in Chouldechova’s case on COMPAS specifically [4]; that compression disproportionately
 43 damages underrepresented groups is likewise documented [9, 3]. We claim neither as new. What is
 44 new is the setting: those results come from full-precision models and workstation tooling, and the
 45 model deployed to a microcontroller is neither. The two interact in a way that matters operationally.
 46 Compression moves parity and calibration in *opposite* directions, so the metric most audits report
 47 improves as the model gets worse, and no toolkit measures this on the compressed artifact inside the
 48 device’s own budget.

49 Contributions:

- 50 1. An on-device audit instrument: a seven-stage pipeline running fairness, uncertainty, and ex-
 51 plainability on the compressed model, emitting a one-page traffic-light card and a machine-
 52 readable manifest.
- 53 2. A measurement showing the calibration/parity tension is group-selective in practice on the
 54 deployed model: the two lenses rank sensitive groups differently on Adult and replicate on
 55 COMPAS, whose favorable direction is inverted (Section 4).
- 56 3. A concrete failure mode for compression-era audits: under pruning, demographic parity
 57 falls roughly 12-fold while per-group calibration error rises roughly 10-fold, so a parity-
 58 only audit ranks the most degraded model as the fairest (Section 5), and the same collapse
 59 replicates on ACSIncome.
- 60 4. Evidence that the uncertainty lens is not redundant with calibration: predictive entropy
 61 ranks the sensitive groups differently from per-group ECE (mean $|\rho| = 0.21$ over the race
 62 cells), so “uncertainty-aware” earns its name rather than restating calibration (Section 6).
- 63 5. A feasibility measurement of the audit itself, which to our knowledge has not been reported
 64 before: peak working set is about 0.6 MB plus 2.3 KB per sample and is constant in test-set
 65 size when streamed (Section 7).

66 2 The TinyAudit instrument

67 TinyAudit takes a fitted model and a labeled test set and produces a one-page audit card plus a JSON
 68 manifest that every stage writes to. The card is generated from the manifest and never hand-edited;
 69 every figure in it is backed by a committed CSV. The stages, in order:

70 **1. Profile.** Parameter count, an analytic FLOPs proxy, peak RAM (via `tracemalloc`), and wall-
 71 clock per sample. **2. Point fairness.** Demographic-parity difference (DP), equalized-odds differ-
 72 ence (EO), and disparate-impact ratio (DI), each with an explicitly pinned favorable direction and
 73 a hypothesis-tested range. **3. Uncertainty.** MC dropout (MLP only), a lightweight perturbation
 74 ensemble, and a QUTE-style early-exit estimator [7]; each returns predictive entropy, variance, and
 75 mutual information. **4. Uncertainty-aware fairness.** Per-group predictive entropy, per-group ex-
 76 pected calibration error (ECE), and a selective-fairness AUC that integrates parity as low-confidence
 77 predictions are progressively abstained. **5. Explainability.** A dependency-free occlusion explainer
 78 reporting per-group feature importance and which features enter one group’s top- k but not another’s.
 79 SHAP and LIME wrappers ship with the package but stay out of the audit path, being costly and
 80 fragile at the edge [18, 15]. The stage is counted in the footprint of Section 7; it yields no fair-
 81 ness result here (Section 8). **6. Compression.** int8 dynamic quantization or magnitude pruning,
 82 applied *before* every metric above, so the audit sees the deployed model. **7. Render.** The one-page,
 83 traffic-light-coded card.

84 The audit’s working set is dominated by per-sample buffers, so it streams in small batches at a peak
 85 RAM independent of test-set size (Section 7). The whole public API is one call:

```
86 from tinyaudit import audit
87 card = audit(model=clf, data=(X_test, y_test), sensitive="sex",
```

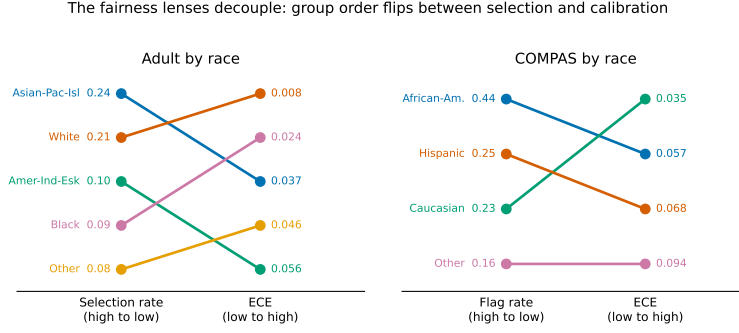


Figure 1: Finding 1. Rank slope charts for Adult (left) and COMPAS (right) by race, uncompressed logistic model, means over 10 seeds. Each line joins a group’s rank by selection/flag rate to its rank by per-group ECE. Crossing lines mean the two lenses order the groups differently.

```
88         methods=["fairness", "uncertainty", "xai"], compression="int8")
89 card.to_pdf("adult_lr_int8.pdf")
```

3 Experimental setup

Datasets. UCI Adult [12] (36,631/12,211 train/test, 101 features), COMPAS [1] (4,629/1,543, 5 features; here the positive outcome is being flagged high-risk, so higher is worse), and Folktables ACSIncome [5] (146,748/48,917, 8 features, 2018 California). Sensitive attributes are sex and race throughout. The first two are the near-universal fairness benchmarks [6]; ACSIncome is the modern replacement for Adult’s 1994 distribution [5].

Models. Scikit-learn estimators on standardized features: logistic regression, an MLP with two hidden layers of 32 units, and a decision tree. Scaling matters, since without it the Adult MLP sits near the majority-class rate (0.54 accuracy) versus 0.84 with it. The scaler is fit on train only and applied outside the model, so compression and perturbation reach the raw weights.

Reproducibility. A single seed flag seeds random, NumPy, torch, and PYTHONHASHSEED. Every headline number is a mean $\pm$ std over seeds 0 to 9. Determinism is guaranteed on Linux/x86; elsewhere we expect last-digit float drift, which the std columns absorb. All models land in published ranges (Adult logistic regression 0.852 ± 0.003 accuracy at ROC-AUC 0.906 ± 0.002 , Adult MLP 0.837 ± 0.003 , COMPAS logistic 0.680 ± 0.006 , ACSIncome MLP 0.808 ± 0.003 ; the full grid is a committed CSV). The fairness results therefore sit on models that work, not on degenerate predictors.

4 Result 1: the two lenses select different groups

Point-prediction fairness asks how often each group receives the favorable outcome. Calibration fairness asks how honest the model’s confidence is per group. The impossibility results [11, 4] say these cannot generally agree; what they do not say is *which* group each lens singles out on a given deployed model, and that is what an auditor actually has to act on. On the uncompressed logistic model, the two lenses rank the sensitive groups differently, and the disagreement is not a uniform offset but a reordering (Figure 1).

Adult by race (five groups). By selection rate Asian-Pacific-Islander (0.240) and White (0.207) rank highest and the smaller groups lowest; calibration reorders them — White is only second-most-selected yet by far the best calibrated (ECE 0.008 ± 0.002 , three to seven times lower than every other group), while American-Indian-Eskimo is worst (0.056 ± 0.019). The smaller groups carry higher, noisier calibration error the selection view never surfaces. **By sex** the disagreement inverts: a large selection gap (Male 0.253 vs Female 0.079) with negligible calibration difference (ECE 0.009 vs 0.014).

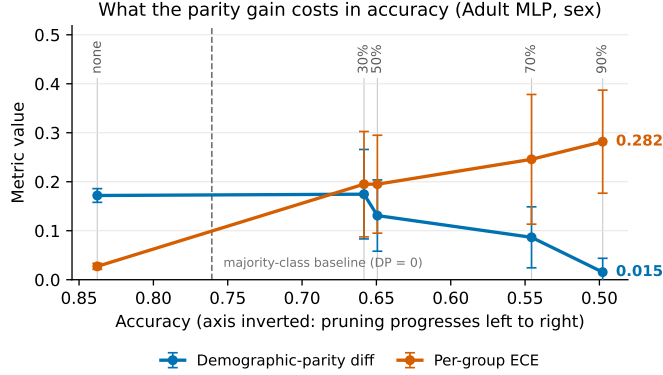


Figure 2: Finding 2. Adult MLP by sex under magnitude pruning. Demographic-parity difference falls while per-group ECE climbs; bands are ± 1 std over 10 seeds. The bands overlap at intermediate sparsities, so the robust claim is the endpoint contrast, not the middle ordering.

121 **COMPAS by race** (positive outcome = flagged high-risk, so higher is worse; groups with $n < 10$
122 dropped). African-American defendants are flagged at roughly twice the rate of the next group
123 (0.441 ± 0.015 vs Hispanic 0.249 ± 0.046), the well-known point disparity. Yet the worst-calibrated
124 group, Other (ECE 0.094 ± 0.029), is flagged *least* (0.164 ± 0.033), and the best calibrated is
125 Caucasian (0.035 ± 0.016). The reordering therefore holds on a second dataset whose favorable
126 direction is inverted, so it is not an Adult-specific label-convention artifact.

127 **Why it matters on-device.** Sensitive labels are rarely available at inference time, so on-device audit
128 triggers lean on confidence scores as a proxy [10]; if the lenses disagree about which group is worst
129 served, a uniform confidence threshold can track the wrong disparity [17, 13]. Reporting both is the
130 cheap fix, and TinyAudit does so by construction.

131 5 Result 2: compression can hide unfairness

132 We prune the Adult MLP to increasing sparsity and re-run the full audit at each level (Figure 2). By
133 sex, the demographic-parity difference falls from 0.172 ± 0.014 (unpruned) to 0.015 ± 0.028 at 90%
134 sparsity, a roughly 12-fold drop, while per-group ECE climbs from 0.027 ± 0.006 to 0.282 ± 0.105 ,
135 a roughly 10-fold rise. By race the shape is the same: DP $0.152 \pm 0.025 \rightarrow 0.022 \pm 0.037$, ECE
136 $0.050 \pm 0.006 \rightarrow 0.314 \pm 0.107$. The model looks fairer on parity precisely as its calibration
137 collapses.

138 The mechanism is degradation toward a near-constant predictor, whose endpoint is textbook: a
139 constant classifier trivially satisfies demographic parity, the standard counterexample for why parity
140 alone is insufficient [16]. Our point is that a routine compression pipeline walks a model toward
141 it gradually and silently while the most commonly reported audit metric improves the whole way,
142 so nothing in a parity-only report distinguishes a model that became fairer from one that stopped
143 predicting. The COMPAS logistic model at 90% pruning (about 5 features) makes it vivid: it predicts
144 one class for everyone and reports DP = 0.000, DI = 1.000, perfect parity by construction — an
145 illustration of the mechanism, not a trend.

146 Two qualifications. Intermediate pruning levels are seed-noisy (parity std 0.06 to 0.10), so the robust
147 claim is the direction between endpoints. And the effect is strongest where pruning actually damages
148 the model: the logistic model barely degrades (DP $0.174 \pm 0.008 \rightarrow 0.126 \pm 0.014$ at 90%) and
149 shows only a mild version of it.

150 The upshot for auditors: an audit that checks only demographic parity, run only after compression,
151 would report the pruned model as the *fairest* in the sweep. An uncertainty-aware audit flags the same
152 model as the most broken.

6 Result 3: the uncertainty lens is not the calibration lens

We call the audit uncertainty-aware, so it is fair to ask whether the uncertainty view adds anything the calibration view does not already give: per-group ECE also reads confidence, so is “uncertainty-aware” just “calibration-aware” renamed? We test it directly. From the same perturbation ensemble we take the pure uncertainty quantities it already produces — per-group predictive entropy, ensemble variance, and mutual information — and ask whether they single out a *different* group than calibration does (Figure 3, Appendix A).

They do. Over 10 seeds the mean absolute Spearman correlation between the entropy ordering and the ECE ordering is 0.21 across the three race cells (Adult logistic -0.40 , Adult MLP $+0.18$, COMPAS logistic -0.06): the two lenses do not rank the groups together. Entropy is not noise either — on Adult it tracks the *selection-rate* ordering (Spearman $\approx +0.75$), so it is its own coherent axis, not a restatement of calibration. On sex (two groups, so the contrast is a disparity gap rather than a correlation) the entropy gap between groups is large exactly where the ECE gap is negligible (Adult 0.154 vs 0.006; Adult MLP 0.155 vs 0.023): uncertainty is loud where calibration is silent. The one cell where the two instead agree (COMPAS by sex, where the ECE gap is the larger of the two) we report rather than hide. The uncertainty view therefore earns its name: a calibration-only audit would rank the groups differently and miss what it surfaces. The Result 2 collapse also replicates on ACSIncome, a third and larger Census benchmark (Appendix B).

7 Feasibility: what the audit itself costs

Audit tooling is normally not profiled at all: the model’s footprint is reported and the auditor’s is assumed free because it runs on a workstation. If the audit is to run where the model runs, that assumption has to be checked. We profile the full three-method audit with `tracemalloc` and report the pipeline’s own peak RAM and wall-clock, not the audited model’s (Table 1, Appendix C).

The working set is dominated by per-sample buffers: the feature matrix, the ensemble’s stacked prediction arrays, and the explainer’s perturbation batch. Fitting the Adult endpoints (625,005 bytes at batch 64 to 10,061,660 bytes at batch 4096) gives

$$\text{peak RAM} \approx 0.6 \text{ MB} + 2.3 \text{ KB} \times \text{batch size}. \quad (1)$$

The structural claim is the one we would defend: an arbitrarily large test set streams through the audit in fixed-size batches at constant peak RAM, so audit cost is set by the batch size the engineer picks, not by how much test data exists.

The absolute constant needs more care. At a 256-row batch the audit peaks at 0.61 to 0.95 MB in under 0.1 s: not within the 32 to 512 KB parts at the low end of the TinyML range, but within reach of the high end, such as Cortex-M7 parts with roughly 1 MB of SRAM. The engineer controls only the per-sample term, not the 0.6 MB fixed cost, so driving that fixed cost below 512 KB is the concrete target this measurement identifies. Two caveats: the audited Adult MLP itself peaks at 6.1 MB, so it stands in for an MCU-class model rather than being one (the 62 KB COMPAS logistic model is the realistic case), and `tracemalloc` excludes the interpreter while counting Python overhead a compiled port would avoid. What survives is the scaling law, not the constant.

8 Limitations and conclusion

Features are standardized before fitting, so the numbers describe the scaled-input regime. The uncertainty ensemble is five lightly-jittered copies of the compressed model, not five retrains, so uncertainty magnitudes are relative. `int8` models’ float weights are unreachable, so their uncertainty columns are blank; decision trees have no dense weights and sit out the compression grid. Footprint numbers are measured on x86, not on a microcontroller. The explainability stage yields no fairness result: the top- k feature divergence between groups is non-monotone under pruning, within seed noise, and does not replicate on COMPAS, so we report that stage as instrument and cost, not evidence. Neither result we do report is theoretically novel [11, 4, 9]; the contribution is measuring them on the model that ships, where those effects prove measurable, group-selective, and actively misleading at under a megabyte per streaming batch: a parity-only audit run after compression ranks the most degraded model in a sweep as the fairest.

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A Uncertainty-vs-calibration rank charts

Figure 3 shows the per-cell rank slope charts underlying Result 3. The main-text numbers (mean $|\rho| = 0.21$ across the race cells; the per-cell Spearman values; the sex disparity contrast) are self-contained; this figure is the visual form of that finding.

Uncertainty is complementary to calibration: entropy and ECE order the groups differently

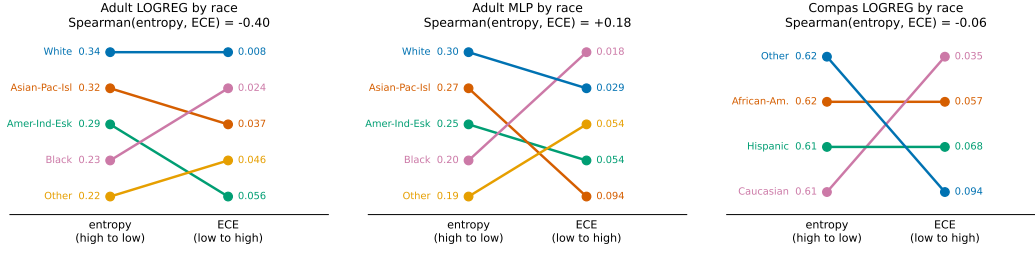


Figure 3: Rank slope charts by race for the three framing-critical cells, uncompressed, means over 10 seeds. Each line joins a group’s rank by predictive entropy to its rank by per-group ECE; crossing lines mean the uncertainty and calibration lenses order the groups differently. Panel titles give the per-cell Spearman rank correlation.

B ACSIncome replication of the compression collapse

The Result 2 effect is not specific to Adult and COMPAS. On ACSIncome (logistic, three seeds, test set subsampled to 8,000 rows for cost), pruning to 90% sparsity drives the demographic-parity difference by sex from 0.034 to 0.000 while accuracy collapses from 0.781 to 0.589 and the positive-prediction rate falls to zero — the “fairest” model in the sweep is again the most broken — and per-group ECE disparity rises from 0.008 to 0.112. The same silent-collapse mechanism appears on a modern, larger Census benchmark.

C Audit pipeline footprint

Table 1: Audit pipeline footprint (the pipeline itself, not the audited model), profiled with tracemalloc over 3 seeds. Peak RAM is governed by the streaming batch size, not the test-set size.

Dataset	Model	Batch	Audit peak RAM (KB)	Audit wall-clock (s)
Adult	logreg	64	610	0.03
Adult	logreg	256	641	0.06
Adult	logreg	1024	2,477	0.11
Adult	logreg	4096	9,826	1.10
Adult	MLP	256	946	0.08
Adult	MLP	4096	9,824	1.28
COMPAS	logreg	256	609	0.01
COMPAS	MLP	1543	1,181	0.05